

PROBABILITY 3: ASSIGNMENT 1

The following problems are all from Manjunath Krishnapur's problem set posted on his webpage.

Problem 1. Suppose $\mathcal{S}$ is a π -system and is further closed under complements ($A \in \mathcal{S}$ implies $A^c \in \mathcal{S}$). Show that $\mathcal{S}$ is an algebra.

Problem 2. Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\mathcal{G} = \{A \in \mathcal{F} : P(A) = 0 \text{ or } 1\}$. Show that $\mathcal{G}$ is a σ -algebra.

Problem 3. (*discussed in class*) Suppose $\sigma(\mathcal{S}) = \mathcal{F}$ and P, Q are two probability measures on $\mathcal{F}$. If $P(A) = Q(A)$ for all $A \in \mathcal{S}$, is it necessarily true that $P(A) = Q(A)$ for all $A \in \mathcal{F}$? If yes, prove it. If not, give a counterexample.

Problem 4. Suppose $\mathcal{F} = \sigma(\mathcal{S})$ and $P(A) \in \{0, \frac{1}{2}, 1\}$ for all $A \in \mathcal{S}$.

- (1) If $\mathcal{S}$ is a π -system, show that $P(A) \in \{0, \frac{1}{2}, 1\}$ for all $A \in \mathcal{F}$.
- (2) If $\mathcal{S}$ is not a π -system, show that it is possible to have $P(A) \notin \{0, \frac{1}{2}, 1\}$ for some $A \in \mathcal{F}$.

Problem 5. Let $\mathcal{F}$ be a σ -algebra on $\mathbb{N}$ that is strictly smaller than the power set. Show that there exist $m \neq n$ such that elements of $\mathcal{F}$ do not separate m and n (i.e., any $A \in \mathcal{F}$ either contains both m, n or neither). Is the same conclusion valid if $\mathbb{N}$ is replaced by any set Ω ?

Problem 6.

- (1) Let $\mathcal{B}$ be the Borel σ -algebra of $\mathbb{R}$. Show that $\mathcal{B}$ contains all closed sets, all compact sets, all intervals of the form $(a, b]$ and $[a, b)$.
- (2) Show that there is a countable family $\mathcal{S}$ of subsets of $\mathbb{R}$ such that $\sigma(\mathcal{S}) = \mathcal{B}_{\mathbb{R}}$.
- (3) Let K be the $1/3$ -Cantor set. Show that $\mu_*(K) = 0$. Here μ_* is the outer measure defined in class.

Problem 7. Show that each of the following collections of subsets of $\mathbb{R}$ generates the same σ -algebra (which we call the Borel σ -algebra).

- (1) $\{(a, b] : a < b\}$.
- (2) $\{[a, b] : a \leq b \text{ and } a, b \in \mathbb{Q}\}$.
- (3) The collection of all open sets.
- (4) The collection of all compact sets.

Problem 8.

- (1) Let X be an arbitrary set. Let $\mathcal{S}$ be the collection of all singletons in Ω . Describe $\sigma(\mathcal{S})$.

- (2) Let $\mathcal{S} = \{(a, b] \cup [-b, -a) : a < b \text{ are real numbers}\}$. Show that $\sigma(\mathcal{S})$ is strictly smaller than the Borel σ -algebra of $\mathbb{R}$.
- (3) Suppose $\mathcal{S}$ is a collection of subsets of X and a, b are two elements of X such that any set in $\mathcal{S}$ either contains a and b both, or contains neither. Let $\mathcal{F} = \sigma(\mathcal{S})$. Show that any set in $\mathcal{F}$ has the same property (either contains both a and b or contains neither).

Problem 9. On $\mathbb{N} = \{1, 2, \dots\}$, let A_p denote the subset of numbers divisible by p . Describe $\sigma(\{A_p : p \text{ is prime}\})$ as explicitly as possible.

Problem 10. If $\mathcal{G} \subseteq \mathcal{F}$ are σ -algebras on Ω and $\mathcal{F}$ is countably generated (i.e., there is a countable collection of sets that generates the σ -algebra), then is it necessarily true that $\mathcal{G}$ is countably generated?

Problem 11. (*discussed in class*) Let $\mathcal{F} = \sigma\{A_i : i \in I\}$ where $A_i, i \in I$, are subsets of Ω . Given $B \in \mathcal{F}$, show that there is a countable subset $J \subseteq I$ such that $B \in \sigma\{A_i : i \in J\}$.

Problem 12. Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measure spaces, i.e., $\mathcal{F}$ and $\mathcal{G}$ are σ -algebras on X and Y respectively. If $T : X \rightarrow Y$ is a function, show that:

- (1) $\{T^{-1}B : B \in \mathcal{G}\}$ is a σ -algebra on X , and
- (2) $\{B \in \mathcal{G} : T^{-1}B \in \mathcal{F}\}$ is a σ -algebra on Y .

Problem 13. Let $A_1, A_2, \dots$ be a finite or countable partition of a non-empty set Ω (i.e., A_i are pairwise disjoint and their union is Ω). What is the σ -algebra generated by the collection of subsets $\{A_n\}$? What is the algebra generated by the same collection of subsets?

More problems

Problem 1*. Let $E = \mathbb{R}^{[0,1]}$ be the space of all functions from $[0, 1]$ to $\mathbb{R}$. Let $\mathcal{F}$ be the cylinder σ -algebra on E (cylinders are sets of the form $\{f \in E : f(t_i) \in B_i, 1 \leq i \leq n\}$ for some $t_1 < \dots < t_n$ and some $B_i \in \mathcal{B}_{\mathbb{R}}$).

- (1) Show that if $A \in \mathcal{F}$, then there is a countable set $\{t_i\} \subseteq [0, 1]$ such that membership in A is determined by the values of a function on the subset $\{t_i\}$.
- (2) Show that $C[0, 1]$ is not a measurable subset of E .

[Remark: The gist is that E is not a good space to model random functions]

Problem 2*. On $C[0, 1]$, show that the Borel σ -algebra and the cylinder σ -algebra are the same. Here Borel σ -algebra is w.r.t. the topology induced by the sup-norm metric. And cylinder sets are those of the form $\{f \in C[0, 1] : f(t_i) \in B_i, 1 \leq i \leq n\}$ for some $0 \leq t_1 < \dots < t_n \leq 1$ and some $B_i \in \mathcal{B}_{\mathbb{R}}$.